PAVAN DHARMA ADAPA

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PROFESSIONAL SUMMARY

Data Analyst with an M.S. in Computer Science & Engineering, **available immediately**. Turns multi-source data into decisions using SQL, Python, Tableau, Power BI, and Advanced Excel. Experienced building end-to-end ETL pipelines, dimensional models, and executive dashboards, and rigorous about the statistics underneath them — complex-survey estimation, experiment design and sequential testing, model calibration, and variance reduction. Portfolio work runs on **real public datasets** at national scale (CDC, CMS, multi-hospital encounter data), with every figure reproducible from the published code.

TECHNICAL SKILLS

Data Analysis & Querying: SQL (joins, CTEs, window functions, query optimization), Python (Pandas, NumPy, SciPy, scikit-learn), R (ggplot2, dplyr), Advanced Excel (pivot tables, lookups, automation).

BI & Visualization: Microsoft Power BI, Tableau, Google Data Studio, Matplotlib, Seaborn — interactive dashboards, KPI reporting, executive-ready visualization.

Databases & Cloud: MySQL, PostgreSQL, SQLite, AWS; connecting BI tools to cloud-hosted sources.

Experimentation & Statistics: A/B test design, power and sample-size analysis, sequential testing (alpha spending, always-valid inference), CUPED variance reduction, sample ratio mismatch detection, hypothesis testing, complex-survey estimation, EDA.

Machine Learning: Logistic regression, Random Forest, XGBoost, gradient boosting, Gaussian Process Regression, MLP, cross-validation, model calibration, decision curve analysis.

Data Processing & Methodologies: ETL/ELT, data cleansing, wrangling, data mining, feature engineering, dimensional modeling, root cause analysis, KPI tracking, data pipelines.

Other Tools: Git, Linux, Selenium, NLP, LLMs, Jupyter.

PROFESSIONAL EXPERIENCE

Graduate Teaching Assistant — Sep 2025 – May 2026

Mississippi State University — Starkville, MS

- Led lab sessions in Python (Pandas, Matplotlib) and R (ggplot2, dplyr) across 6 data science courses, demonstrating how to clean, transform, and visualize datasets and communicate insight to non-technical audiences.
- Analyzed student performance data using Advanced Excel, SQL, and Python to track KPIs such as assignment completion, grade distribution, and retention, enabling instructors to identify at-risk students early.
- Designed and evaluated coding assignments simulating real analyst workflows — preprocessing, feature engineering, EDA, model evaluation — improving project completion speed by **25%**.

Graduate Research Analyst — Aug 2024 – Aug 2025

Mississippi State University — Starkville, MS

- Analyzed **900,000+** epidemiological records using Python, SQL, and EDA to identify key environmental drivers of disease transmission, producing data-backed recommendations for stakeholders in CSE and Veterinary Medicine.
- Built an automated ETL pipeline in Python and Selenium to collect, cleanse, and standardize multi-source datasets, reducing metadata processing time **95%** — from hours to under 10 minutes per dataset.
- Developed and evaluated ensemble predictive models (Gaussian Process Regression, Random Forest, XGBoost) to forecast disease outbreaks, achieving **87% predictive accuracy** and establishing baseline KPIs for model performance tracking.
- Automated feature-engineering and data-cleansing workflows, resolving **35%** of dataset inconsistencies and reducing manual analysis time **50%**.
- Created correlation heatmaps and statistical visualizations surfacing **8 key influencing factors**, packaging results into presentation-ready dashboards for cross-functional faculty collaborators.
- Simulated 10,000+ outbreak scenarios using statistical analysis and hypothesis testing, contributing results to an interdisciplinary research manuscript.

Marketing Data Analyst Intern — May 2023 – Dec 2023

Viral Fission

- Developed digital marketing analytics and reporting systems tracking audience engagement, funnel performance, and campaign ROI for youth-focused brand campaigns with clients including Coca-Cola, ITC, and Mahindra.
- Built automated Power BI and GA4 dashboards that reduced manual reporting time **50%** and enabled real-time monitoring of reach, engagement rate, click-through rate, and conversion KPIs.
- Applied SEO optimization, A/B testing, and UX insights from GA4 and social analytics to refine content and posting strategy, contributing to a **25% increase** in key audience-interaction metrics, and translated event-tracking data into recommendations on content, channel mix, and campaign timing for brand and marketing leadership.

Data Analytics Intern — Aug 2022 – Dec 2022

InfraBIM Techno Solutions

- Led analytics and reporting-automation initiatives on engineering and operations data using Power BI, SQL, and Google Colab, improving visibility into project performance and stakeholder KPIs.

- Developed KPI dashboards, ETL pipelines, and automated reporting that reduced manual reporting effort by ~**30%** and enabled faster data-driven decision-making.
- Conducted EDA to surface patterns, trends, and recurring data-quality issues, and evaluated model performance across experiments — summarizing accuracy and error metrics in analysis-ready tables to guide model selection.

ANALYTICS PROJECTS

Experimentation & A/B Test Analysis Toolkit · Estimator validation against known ground truth ·
github.com/adapapavandharma/experimentation-toolkit

- Quantified the peeking problem by simulation: stopping at the first significant look across 10 interim analyses drives the false positive rate to **18.5% against a nominal 5%**; implemented O'Brien–Fleming alpha spending and always-valid mSPRT bounds and measured what each costs in power.
- Implemented **CUPED** variance reduction matching the ρ^2 prediction to within a tenth of a point — at $\rho = 0.7$ worth **1.96× the traffic at no cost** — with unbiasedness verified across 300 replications.
- Built sample-ratio-mismatch detection and demonstrated a global SRM check **passing cleanly** on an experiment whose segments are catastrophically imbalanced, establishing per-segment SRM as the necessary control.

Claims Denials & Appeals Analytics · CMS Transparency in Coverage — 3-year national panel ·
github.com/adapapavandharma/denials-analytics

- Built a 15,545 plan-year panel covering **126.9 million denied claims** across every Qualified Health Plan on the federal Marketplace, 2021–2023.
- Established that only **0.168% of denials are ever appealed** while **40.1% of appeals succeed**, identifying capacity to file — not claim merit — as the binding constraint on denial recovery.
- Showed the payer-reported reason taxonomy cannot support root-cause work (**"Other" is 71.7%** of categorised denials and growing), and that denial rate is an organisational property: **6.0% to 48.3%** across issuers with 100,000+ claims.

Emergency Department Throughput Analytics · CDC/NCHS NHAMCS — real national survey microdata ·
github.com/adapapavandharma/ed-throughput-analytics

- Analyzed **16,025 real emergency department visits**, survey-weighted to the **155.4M** US ED visits of 2022; the pipeline self-validates by reproducing CDC's published national total exactly or refusing to run.
- Implemented the **ultimate-cluster variance estimator with Taylor linearisation** CDC prescribes for this stratified multi-stage sample; design-correct inference showed the hypothesised peak-hour effect does **not** clear significance at $\alpha = 0.01$ — a negative result reported rather than discarded.
- Found between-hospital variation in median door-to-provider time (2–74 min) to be **3.9× larger** than time-of-day variation, and quantified boarding at **48.6M ED bay-hours lost annually** as the recoverable capacity item.

30-Day Readmission Risk & Care-Management Registry · 99,343 real inpatient encounters, 130 hospitals ·
github.com/adapapavandharma/readmission-risk

- Built a gradient-boosted readmission model (**AUC 0.664**) on real multi-hospital encounter data; top-quintile targeting captures **37.7% of all readmissions** at **1.89× enrichment**, number-needed-to-screen 4.8, delivered as a capacity-derived tiered risk board.
- Benchmarked against a single-variable clinical rule to isolate marginal value — prior inpatient count alone reaches 32.2% recall, so the model buys **+5.5 points**, materially less than an AUC comparison implies.
- Showed balanced class weighting inflates mean predicted risk to **46% against an observed 11%** while leaving AUC unchanged, establishing calibration rather than discrimination as the constraint on deployability.

Multimodal LLM Evaluation (LLaVA-1.5) · Analytics & experimentation — academic research

- Designed evaluation pipelines analysing cross-lingual performance of multimodal LLMs — NLP techniques, data-contamination auditing, and statistical comparison across benchmarks — aggregating experiment metrics in Python and SQL and presenting findings to academic stakeholders.

EDUCATION

M.S., Computer Science & Engineering Graduated May 2026
Mississippi State University — Mississippi State, MS
Coursework: Advanced Machine Learning, Artificial Intelligence, Algorithms, Parallel Algorithms, Software Engineering, Data Analytics, Advanced Cyber Operations.

B.Tech, Computer Science & Engineering Graduated May 2024
Malla Reddy University — Hyderabad, India
Coursework: Data Structures & Algorithms, DBMS, Statistics & Probability, Machine Learning, Operating Systems, Computer Networks, OOP.

PUBLICATION (IN PROGRESS)

Computational Tools for Modeling and Predicting Respiratory Disease Spread in Commercial Poultry Production
Zhang Li, Pavan Dharma Adapa, Sai Harika Gade, Saikanth Ratnavale, Navicky Michael.

- Designed forecasting frameworks synthesizing compartmental (SIR/SEIR) and machine-learning models to predict respiratory disease transmission at scale, and assessed Random Forest and deep-learning methods for identifying risk factors and detecting anomalies in surveillance data.